Collimator VCurve — Quick Manual (v5.7.4)

Objective flange focal distance (FFD) measurement for double-pass autocollimator benches. A USB camera replaces your eye at the collimator: the app measures image sharpness (Tenengrad metric) in the R, G and B channels while you step the lens through focus, then fits a parabola to each channel's curve and reports the vertex — the true best-focus position — with its uncertainty. **The G (green) channel is the reference measurement; R–G and B–G show the lens's secondary spectrum.**

Works the same on Windows, macOS and Linux. Touchscreen-friendly.

1. Install & first run

OS	Steps
Windows	Run `CollimatorVCurve-setup.exe` and follow the wizard (no admin needed). Launch from the Start Menu. If SmartScreen appears: *More info → Run anyway*.
macOS	Open the `.dmg`, drag the app to Applications. First launch: **right-click** → **Open** (Gatekeeper). Allow camera access when asked.
Linux	Ubuntu/Debian: install `CollimatorVCurve-amd64.deb` (double-click or `sudo apt install ./CollimatorVCurve-amd64.deb`), launch from the menu. Other distros: `chmod +x CollimatorVCurve-linux-x86_64`, then run it.

Connect the USB camera **before** starting the app. The app opens as a maximized window filling the screen, finds the camera automatically and remembers your last camera, language and unit (mm/µm) between sessions. A fresh install starts **in English**; switch to Portuguese any time in SETTINGS → *language*.

2. The screen

- **Live view** (left): the camera image in the selected channel's color. Drag a rectangle to set the **ROI** — keep it on the reticle image. The ROI is **remembered between sessions**, so you do not redraw it every time you open the app; if it no longer fits the current frame (you changed camera or resolution) it returns to the centre. The sparkline at the bottom shows the live focus metric with its **RMS** readout and a dashed **peak-hold** line marking the best value seen.
- **Zoom** (view only — the measurement always uses the raw image): pinch (= Ctrl + wheel), the **ZOOM – / + / MAX** row, `+` / `–` keys, or **double-tap**. Centered on the ROI; step, max (5×/10×) and gestures configurable in CONFIG. The third zoom button says what the tap will do: it reads **MAX** while you are at 1×, and **1x** once you are zoomed in.
- **RESET %**: sets a new 100% reference for the live readout — after changing the aperture, for example — without clearing the points already taken.
- **UNDO**: removes the last point you took. Typed the micrometer reading wrong? Tap UNDO and take it again — no need to CLEAR and sweep everything from scratch. Key `u`. Grey when there is nothing to undo.
- **Buttons**: POINT · FIT · UNDO · CLEAR · CONFIG · PHOTO, plus the ZOOM row, RESET %, the **?** (help) button and **CLOSE**. The app picks the layout by itself: a side panel on 16:9 screens, or a band under the video on 3:2 / 16:10 screens (which gives the video more room).
- **CONFIG**: channel (G is the official measurement channel), language, camera, resolution, output folder, zoom step & max zoom, and exposure (LOCKED / AUTO).
- **CLOSE**: tap twice to quit (first tap asks "QUIT?"). ESC or `q` also work.
- **? (help)**: features, shortcuts and where to report bugs or suggest improvements.

3. Measuring a V-curve

1. Set up the bench as usual: lens under test + mirror at the flange plane; reticle visible and roughly focused.

2. Select the **G channel** (button CHANNEL or key `g`).
3. Place the ROI over the reticle image. Draw the ROI by tapping the target and dragging outward. Set the aperture first, then RESET %, then focus. Zoom (- + MAX) and RESET % live next to the buttons. CONFIG holds channel, language, camera, resolution, folder, zoom and exposure. All output files, including the log, go to the output folder.
4. Move the lens/stage to one side of best focus. For each step:
   - Press **POINT** (or space bar).
   - Type the **micrometer position** on the on-screen pad. It opens with the **previous value selected** (shown inverted): just type to replace it, or press OK to repeat it. The `–` key toggles a negative sign; the **mm | um** toggle at the top switches the input unit (µm needs no decimal point: type `550` instead of `0.55`). Press **OK**.
5. Take **at least 5 points** (FIT unlocks at 5; 7–11 is better), bracketing the peak — go *through* best focus, don't stop at it.
6. Press **FIT**. The app saves a CSV and shows, per channel: `vertex ± sigma (points used)` and the spectral offsets `R−G` / `B−G` in µm. **Use the G vertex** as the FFD result. A **mirror tilt** line is also shown: the app tracks how the image walks sideways during the scan (the mirror rotates with the micrometer spindle, 0.5 mm/turn, so a tilted mirror traces a spiral). A large value (or a growing rms) means the mirror needs squaring.

Notes: the fit uses points at ≥60% of the peak metric; the app flushes camera buffer frames before each POINT (anti-lag), and camera auto-exposure/WB/focus are disabled automatically. Approach each position from the same direction to avoid micrometer backlash. Do not mix different cameras, ROIs or resolutions inside one scan (the app clears points when resolution changes). FIT flags a scan invalid (vertex outside the sweep, or one side with under 2 points). This check now looks only at the points that actually entered the parabola — the ones at 60% of the peak or above — so a scan that only covers one flank is called invalid instead of quietly passing. If the camera drops resolution, POINT blocks — re-tap the resolution in SETTINGS. If the camera stops answering (a loose USB cable is the usual cause), the video dims and says *reconnecting…* while the app brings it back on its own — it also re-scans every input, because a USB device often comes back on a different index. If it does not return, open SETTINGS → **CAM** and tap the camera again: that always reopens it.

4. Cameras

The **CAM** button inside SETTINGS opens the selector: each input appears as a card with a **live thumbnail**, index, resolution and name. Tap to switch. *SCAN AGAIN* re-probes; the chosen camera is remembered for the next session. On Windows, the camera's **driver panel** (exposure, gain…) is inside SETTINGS — or key `o`.

5. Settings

SETTINGS holds: the **camera resolution** (720p / 1080p default / MAX — higher sharpens focus discrimination; changing it clears the points; the top-right badge shows actual resolution and fps); the camera **DRIVER** panel (Windows only); the **photos & CSV folder** — *CHANGE* picks a new one, *OPEN* shows it in your file manager; the **zoom step** (smooth 5% — the default — 2×, or straight to MAX) and the **max zoom** (5× default, or 10×); and two gesture switches: *double-tap = MAX zoom*, and *plain scroll zooms* without Ctrl (on by default on macOS, where two-finger scrolling is the natural gesture). Everything is remembered between sessions.

6. Updates & reporting

On startup the app checks for a new version (needs internet) and only asks: **UPDATE NOW** downloads and installs it silently, then reopens the app by itself — no wizard, no clicks. **LATER** dismisses. If the automatic update is not possible (offline, Linux `.deb` install), the download page opens instead. You can also tap **CHECK FOR UPDATE** inside **?** — if it fails, it now names the cause (no internet, something blocking the app, a certificate problem, or the server refusing) and prints the technical reason right on that screen. To report a bug or request a feature, tap **SEND REPORT** in **?**: type your message (on-screen or physical keyboard) and send — it goes straight to the support inbox. **You can also just tap SEND without typing anything.** The report always carries the app log (`colimador_log.txt`), your settings, and the DIAG recording of the session if you made one; the screen lists what goes with it. That is usually enough for support to find the problem without asking you for files. Once it goes through, the screen turns into **REPORT SENT** with a big green confirmation and a single **OK** button to close it, so you never have to wonder whether it went. Offline, an E-MAIL button appears as fallback, with the last lines of the log and the path of the files so you can attach them.

7. Files

All output — CSV, photos and the DIAG recordings — goes to one output folder: `Documents/colimador/` by default, or the folder you pick in CONFIG → **CHANGE** (**OPEN** shows it in your file manager). The app's own log, `colimador_log.txt`, lives there too.

File	Content
`vcurve_YYYYMMDD_HHMMSS_WxH.csv`	raw points: `pos_mm, tenengrad_R, G, B, cx_px, cy_px` (positions always in **mm**; the centroid columns feed the mirror-tilt fit; the filename records the camera resolution used)
`snap_*.png`	screen snapshots (PHOTO button)
`diag_YYYYMMDD_HHMMSS.csv`	120 s diagnostic recording (DIAG): ROI mean/stdev, gradients, exposure/gain, camera fps
`colimador_log.txt`	app log — camera opens, warnings, crash traces

`config.json` (language, unit, last camera, folder, zoom & gesture options) lives in the OS app-data folder, not here.

8. Keyboard shortcuts

`space` POINT · `f` FIT · `g r b w` channel · `c` clear points · `+ -` zoom · `o` driver (Win) · `ESC` / `q` quit · keypad: digits, `-`, `.`, Backspace, Enter = OK.

9. Troubleshooting

- **"No camera found"** — plug the USB camera and tap the screen to rescan (CLOSE button quits).
- **Black or dark image** — open DRIVER (Windows) and raise exposure; check the collimator lamp.
- **"fit failed"** — the points don't bracket a peak: add points on both sides of best focus, or widen the scan range.
- **Numbers look 1000× off** — check the mm | um toggle on the keypad.

Collimator VCurve v5.7.4 — 2olhares (Lucas Werneck). The CSV is always in mm. Minimum 5 points per fit. G channel is the official measurement.